

Classification with iris dataset

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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix

iris = load_iris()
X=iris.data
y=iris.target

print(X.shape)
print(y.shape)

(150, 4)
(150,)

print(type(y))
<class 'numpy.ndarray'>

print(iris.feature_names)
print(iris.target_names)

['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal
width (cm)']
['setosa' 'versicolor' 'virginica']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

print(y_test.shape)

(30,)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)

KNeighborsClassifier()

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

Accuracy: 1.0

print(confusion_matrix(y_test, y_pred))

[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```